INCH-POUND

MIL-L-14576F
24 March 1999
SUPERSEDING
MIL-L-145755
5 February 1985

MILITARY SPECIFICATION

LATHE , ARMATURE AND UNDERCUTTER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of bench mounted lathe designed for use in commutator turning and mica undercutting of automotive generator and starter motor armatures.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1. 1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are-those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the-solicitation (see 6.2.1).

SPECIFICATIONS

FEDERAL

W-C-596/66 - Connector, Plug, Electrical, General Purpose, Genera 1 Grade, Grounding, 2 Pole, 3 Wire, 20 Amperes, 250 Volts, 50/60 Hertz.
PPP-B-601 - Box, Wind, Cleated-Plywood.

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
ML-T-152 - Treatment, Moisture and Fungus-Resistant, of Communica-

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.

	tions, Electronic, and Associated Electrical Equipment
MIL-V-173	varnish, Mositue-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-M-1805	Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28	Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-454	Standard General Requirements for Electronic Equipment, Requirement 4.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1 186	Cushioning, anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.
MIL-STD- 1190	Minimum Guidelines for Level C Preservation, Packaging and Marking.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US DEPARTMENT OF LABOR

29 CFR 1910	Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents-, U.S. Government Printing Office, Washington, DC 20402-0001.)

* 2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.1).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1	- Surface Texture.

(Applications for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERCIAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.

ASTM E330 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ' ASSOCIATION (NEMA)

NEMA ICS 1 General Standards for Industrial Control and System

NEPMA MG-1 - Motors and Generators.

(Applications for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 National Electrical Code

(Applications for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02259.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards)) the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. RSQUIREMENT'S

3.1 First article. When specified (see 6.2.1), a sample shall be subject to first article inspection (see 6.4) in accordance with 4.3. The approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The lathe shall be new and of the manufacturer's latest design which shall meet the requirements specified herein. The lathe shall be bench mounted and shall consist, essentially, of a bed, electric motor, headstock, tailstock, tool carriage, and mica undercutter. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the lathe or any lathe component.

3.2.1 Measurement system. The US customary system of units (US) or the International System of Units (SI) may be used in the design and construction of the machine. In this specification all measurements, dimensions, sizes, and capacities are given in the US customs-y system of units (US). These measurements may be converted to the SI through the use of the conversion factors and methods specified in ASTM E380.

Source: https://assist.dla.mil -- Downloaded: 202640-18T03:27Z
Check the source to verify that this is the currentversion before use.

3.2.2 Measuring and indicating devices. Unless otherwise specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and SI systems of measurement. When a dual US and SI system is furnished, feed dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 5.2. 1). Regardless of' the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practices, shall be suitable for the intended purpose in the design and fabrication of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would affect the performance, maintainability, reliability, durability, or longevity of the individual components or the end item equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Raw materials. The manufacturer may use reclaimed materials for fabricating new parts. Reclaimed materials shall be reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the raw materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

* 3.4 Fabrication. The end item of equipment shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the end item of equipment, to its maximum rating and capacity, without permanent distortion.

* 3.4.1 Castings and forgings. Finished castings and forgings shall be free of fins, scale, inclusions, cold shuts, voids, cracks, thermal ruptures, laps, folds, mismatching, or other defects. No processes such as welding, peening, plugging, soldering, or pasting shall be used for repairing, reworking, or improving cosmetic appearance.

* 3.4.2 Welding, brazing, and soldering. All welding, brazing, and soldering shall be free of defects and of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. Defects shall include but are not limited to such things as bubbles, cracks, gouges, holes, and insufficient materials used. All flux or other such materials used during these operations shall be thoroughly removed (cleaned), from the part(s) immediately upon completion of the operation. These operations shall not be employed as repair measures for defective parts.

* 3.4.3 Fastening Devices. Fastening devices and methods shall be chosen to serve the need while providing necessary adjustability for service, maintenance or repair. Fasteners shall be installed to prevent loss of tightness and shall not loosen in service.

3.4.4 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in soldering operations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.

* 3.4.5 Threads. All machined threads shall conform to FED-STD-H28 and the applicable detailed standards referenced therein.

3.4.5 Lubrications. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provied with standard means for lubrication. Oil holes, grease fittings, and filler-caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

* 3.4.7 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide easy access to guarded parts. The lathe shall comply with the general safety and health requirements promulgated under 29CFR 1910 that are applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.5).

* 3.4.8 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective Parts between end items of the same model. All replacement parts shall be 'interchangeable with their respective original parts, and fit/function with mating components.

* 3.4.9 Surfaces. Surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials.

3.5 performance and product characteristics.

3.5.1 Commutator turning. With an armature shaft supported by headstock and tailstock chucks, the lathe shall be capable of machining the commutator to an accuracy within 0.0005 inch total indicator reading runout and a taper of not more than 0.005 inch per linear inch of the turned cylindrical surface. The machined surface shall have a roughness height value of not more than 63 microinches, arithmetical average (aa), in accordance with ANSI B46.1.

3.5.2 Mica undercutting. With the armature supported as in 3.5.1, and the undercutter equipped with a saw(s), the device shall be capable of undercutting the mica between all bars of the commutator to a depth of 0.0313 $\pm$ 0.0050 inch. The undercut shall be parallel to the centerline within $\pm$ 0.005 inch for its full length and width.

3.5.3 Capacity. The lathe shall have at least a six inch swing over the bed and undercutter carriage. It shall be capable of machining and undercutting commutators up to, and including, 3.3125 inches in diameter and 2.5 inches in length. The lathe shall accommodate armatures up to, and including, 13 inches in length, shaft included.

3.6 Components

3.6.1 Lathe bed. The lathe bed shall be sufficiently strong and rigid, with machined and finished bearing surfaces, to assure the accuracies specified herein. When specified (see 6.2.1), a sub-base shall be provided to permit leveling of the lathe on bench.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.

3.6.1.1 Lathe bed ways. The lathe bed shall assure correct alignment of the headstock with the tailstock. The ways shall be of open-end construction to permit removal of the headstock and tailstock assemblies.

3.6.2 Headstock. The headstock shall be capable of supporting the commutator and of the armature being reconditioned. The base of the headstock shall be machined to fit the bed ways. Provision shall be made for clamping the headstock to the lathe bed.

3.6.2.1 Headstock center hole. The headstock center hole shall be not less than 1.0625 inches in diameter. Means shall be provided for locking thrust plugs and lathe centers in the center holes of both the headstock and tailstock.

3.6.3 Tailstock. The tailstock shall be capable of supporting the end of the armature opposite the commutator. The base of the tailstock shall be machined to fit the bed ways. Means shall be provided to clamp the tailstock in position on the lathe bed.

3.6.3.1 Headstock, tailstock alignment. Vertical and horizontal alignment of the headstock center with the tailstock center shall be within 0.002 inch total indicator reading (tir).

3.6.4 Tool carriage. The tool carriage shall be located at the headstock end of the lathe. Longitudinal feed shall be provided by a lead screw which shall be actuated by a ball crank handle or a handwheel.

3.6.4.1 Tool carriage guides. The tool carriage guides shall provide support to the tool carriage for the entire range of travel.

3.6.4.2 Tool holder. The tool carriage shall be equipped with a tool holder and a cutting bit ground for commutator turning. The tool holder shall be provided with means for adjusting the depth of cut and a locking device for locking the tool at any setting. The depth of cut shall be continuously adjustable over its range. Adjustment shall be provided by a graduated thumbscrew or feed wheel.

3.6.5 Chucks Both the headstock and tailstock shall be equipped with self-centering chucks with a minimum range of 0.2031 inch to 1.0625 inches diameter. Chucks shall be fitted with replaceable jaws and shall support the armature by means of' contact between the armature shaft bearing areas and the chuck jaws.

3.6.6 Thrust plugs. The lathe shall be provided with thrust plugs for insertion in the headstock and tailstock centerholes.

3.6.7 Lathe centers. Hardened steel headstock and tailstock centers shall be furnished with each lathe. The lathe centers shall be capable of being easily installed and removed.

3.6.8 Drive. Unless otherwise specified (see 6.2.1), the drive for turning commutators shall consist of a belt encircling the core of the armature and the drive motor pulley.

3.6.9 Mica undercutter. A mica undercutter attachment shall be provided and shall include a complete set of saws of the sizes specified in 3.9. The attachment shall be either a self-powered motorized unit or a belt driven unit, as specified (see 6.2.1). The attachment shall be the manufacturer's standard equipment and shall be capable of being mounted on the back of the lathe bed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.

3.6.9-1 Undercutter depth adjustment. The depth of the cut shall be provided with means of adjustment capable of being locked at any setting.

3.5.9.2 Undercutter feed lever. The longitudinal feed of the undercutter shall be provided, and controlled, by a hand lever designed to offer a mechanical advantage of not less than 3 to 1.

3.7 Electrical components. There shall be no component part used in any application where the requirements exceed the part manufacturer's rating for that part .

3.7.1 Control switches. The drive motor shall be equipped with an on-off switch. The undercutter motor, when included, shall also be equipped with an on-off switch. Switches shall conform to NEMA ICS-1.

3.7.2 Power supply cord. Unless otherwise specified (see 6.2.1), the drive motor, and undercutter motor when provided, shall be furnished with not less than 8 feet of 3-conductor flexible electrical cord, No. 18 American Wire Gage, National Electrical Code type SO, STO, or SJO, heavy duty conforming to NFPA 70 The cord shall be provided with a three-prong grounding plug conforming to W-C-596/66. A mating three-to-two parallel prong adapter with a grounding lead shall be furnished to adapt the plug for use with a two-prong outlet.

* 3.7.3 Armature drive motor. Unless otherwise specified (see 6.2.1), the drive motor shall be not less than 1/4 horsepower, squirrel-cage induction-type motor conforming to NEMA MG-1.

* 3.7.3.1 Undercutter motor. When provided, the undercutter motor shall be a squirrel-cage, induction-type motor conforming to NEMA MC-1. The undercutter motor shall operate on the same current as the armature drive motor.

3.7.4 voltage and current. Unless otherwise specified (see 6.2.1), the lathe shall operate on nominal 115/230 volts (initially wired for 115 volts), single phase, 60 hertz, alternating current.

3.7.5 Motor bracket. A hinged motor bracket shall be provided for mounting the drive motor. The bracket shall be attached to the back of the lathe bed. Means of vertical and longitudinal adjustment of the bracket shall be provided to control belt tension and alignment.

* 3.8 Fungus resistance. Electrical and electronic components and electrical insulation not otherwise specified herein skill be fungus inert, in accordance with requirement 4 of MIL-STD-454. Components of electrical circuitry that are not weather resistant to moisture and fungus growth shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.9 Accessories. Unless otherwise specified (see 6.2.1), in addition to the tool bit specified in 3.6.4.2 and the undercutting saw specified in 3.6.9, the following accessories shall be furnished with each lathe:

1 - Extra tool bit, ground for commutator turning.
6 - Extra undercutter saws, high speed steel 0.3125 inch diameter, 0.125 inch arbor hole consisting of two (each) saws of the following thicknesses: 0.015, 0.020, and 0.025 inch.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:27Z
Check the source to verify that this is the current version before use.

3.10 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least. one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel) shall be applied.

3.11 Product identification. The lathe shall be marked for product identification in accordance with MIL-STP-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.12 Workmanship. Standards of workmanship shall assure that the lathe shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and test) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified in the contract, first article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all the tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.

(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).

* 4.5 Product examination Visually, dimensionally, and manually examine each item to determine conformance with the requirements of 3.2 through 3.4.9, 3.5.3 through 3.6.3, and 3.6.4 through 3.12. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to identified specified standards, adequacy of markings, proper cleaning, and freedom from defects. Dimensional examination includes measuring dimensions as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practial point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the item and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.4.3), shall be cause for rejection.

4.6 Tests.

4.6.1 Operational test. The lathe shall be operated at no load to determine proper operation of all controls, motors, adjusting mechanisms, and other accessories.

* 4.6.2 Alignment test. A cylindrical test mandrel, no less than 10 inches long, shall be mounted in the headstock and tailstock chucks. Dial indicators shall be mounted to the saw fixture and tool post. The fixture and post shall then be traversed a distance of no less than 2.5 inches along the mandrel to measure the vertical and horizontal alignment. A variation in either horizontal or vertical alignment of 0.002 inch or greater shall be cause for rejection (see 3.6.3. I).

* 4.6.3 Commutator turning test. The lathe must have passed the tests of 4.5.1 and 4.6.2 before performing this test. The test shall be performed on a motor or generator armature having a commutator no less than one inch in diameter, nor less than an inch in length. The operator shall take a series of cuts to reduce the commutator diameter by at least 0.025 inch over the full length of the commutator. The turned surface finish shall be not more than 63 microinches arithmetic average (aa), in accordance with ANSI B46.1. Variation in concentricity of the finished commutator shall not exceed 0.0005 inch total indicator reading. Taper shall not exceed 0.005 inch per linear inch of the turned surface (see 3.5.1).

4.6.4 Mica undercutting test. Following completion of the test in 4.6.3, the sample lathes shall be subjected to a mica undercutting test. While the armature is still in the lathe, the operator shall use the mica undercutting device to undercut the mica between the commutator bars to a depth of 0.0313 $\pm$ 0.005 inch. The undercut shall be uniform in depth within + 0.005 inch for its full length and width (see 3.5.2).

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of' Section 5.

5. PACKAGING

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.

5.1 First article pack . When specified, the supplier shall furnish one complete pack for first article packaging inspection. The first article may be either a preproduction pack or an initial production pack which conforms to the requirements stated herein. In either case, the approved first article and the production packs shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with (IAW) the requirements stated herein.

5.2 Disassembly. When practical, for the protection of components, attachments and accessories against damage, pilferage or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassembled parts shall be preserved, anchored, braced, blocked, and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

5.3 Disassembled parts marking. Disassembled parts shall be marked, when necessary with instructions, to facilitate reassembly. Removed parts and mating parts on the item shall be marked by stenciling, on the part, or by use of tags. Tags shall be waterproofed and ink used (on parts or tags) shall be waterproof. Tags shall be firmly secured to each individual part.

5.4 General requirements. Component parts of machinery, accessories, and maintenance tools shall be cleaned, dried, preserved, packaged, and packed as specified herein. Parts and assemblies which are not covered herein shall be preserved and packaged utilizing the materials and methods specified for similar parts and assembles described herein or IAW MIL-M-18058. Cleaning and drying procedures, and preservatives and methods shall conform to the guidelines specified herein, and in MIL-P-116.

5.5 Preservation. Preservation shall be Level A or C, as specified (see 6.2.1).

5.5.1 Level A.

5.5.1.1 Cleaning and drying. Thorough cleaning and drying of all bare ferrous metal surfaces is essential before the application of a preservative. Care must be taken to limit the handling of cleaned parts prior to the assembly of new machinery. Where practicable, processors and assemblers shall wear rubber or plastic gloves while handling cleaned parts.

5.5.1.2 Internal surfaces. All ferrous metal surfaces of parts which will be utilized internally in the assembly of new machinery shall be solvent cleaned followed by fingerprint remover and wiped or drain dried. In the interim, to preclude contamination from forming or cleaned surfaces prior to the assembly operation, parts shall be coated with a light weight preservative oil conforming to P-9. To prevent dust from accumulating on cleaned parts, cover with a sheet of chemically neutral paper.

5.5.1.3 External surfaces. All accessories and attachments such as chucks, faceplates, steady rests, toolholders, abrasive wheels, etc., shall be removed from the assembly prior to the external cleaning operation. All external, critical ferrous metal surfaces, fit or function, of the machine shall be solvent cleaned followed by fingerprint remover and wiped or drain dried. Noncritical ferrous metal surfaces shall be solvent cleaned and wiped or drain dried. Nonmetallic composites and painted metal surfaces shall be cleaned by wiping with a clean, dry

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.

cloth. Care must be taken to prevent cleaning solvents from coming in contact with nonmetallic components. When cleaned surfaces of the machine show signs of rusting before application of the final preservative, reprocessing the rusting surface shall be confined to that area of the assembly. The solvent used for the final wiping operation shall contain a small amount, approximately 5 percent by volume, preservative oil conforming to P-10 type 10, grade 30. It is extremely important that elapsed time between these operations be held to an absolute minimum.

5.5.1.4 Disassembled components. Disassembled components detached from the basic machine shall be cleaned and dried as specified herein for like items.

5.5.1.5 Machine tools and accessories. All such tools and accessories shall be clean and free from contaminants. Preservatives shall be applied to provide protection from corrosion during shipment and storage. All preserved items shall be wrapped with greaseproof barrier material. Critical surface items, i.e., parallels, etc., shall also be wrapped with no less than 1/2 inch uncompressed cushioning material.

5.5.1.6 Electrical equipment. Electrical equipment shall be clean and dry. Preservatives shall be applied as required to prevent corrosion to moving or other ferrous metal parts, but shall not come into contact with electrical connections, windings, etc. Openings shall be sealed with cap-plugs or with waterproof, pressure sensitive tape to provide protection against entry of any foreign materials.

5.5.1.7 Lubricating systems. Vent openings in the lubrication systems shall be temporarily capped or plugged to prevent loss of lubricants during shipment. Tags, warning that "ALL TEMPORARY CAPS OR PLUGS MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM", shall be attached to all temporary caps or plugs intended to be vented to the atmosphere during operation. These warnings shall also be included in the instruction manual. In the event a lubricant reservoir cannot be covered and sea led, it shall be appropriately preserved and the lubricant shall be provided in a sealed container and shipped with each machine. A warning tag shall be placed in a conspicuous place near the filling area, warning the user to “INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE EQUIPMENT". The operator’ s manuals shall also reflect the above warning and procedure.

5.5.1.8 Maintenance and hand tools. Maintenance and hand tools such as screwdrivers, hammers, chisels, etc. , shall be clean and free from contaminants. Sharp edges such as cutting edges shall be protected by a fiberboard shield placed on the edges and taped in place. Tools shall be placed in a fiberboard box and secured from movement so as to prevent damage to the tools or other items.

5.5.1.9 High-speed spindle bearings. Fill the reservoir to normal operating level with manufacturer’s operating lubricant. If the system is of such design that the oil will leak out during shipment or storage, the system shall be preserved by filling with preservative oil P-10, type I, grade 30, and drained. Operational lubricant shall be furnished.

5.5.1.10 Grease fittings. Charge all grease fittings with preservative compound conforming to ?-11.

5.5.1.11 Nonlubricated interior surfaces. Surfaces of internal gears which do not run in lubricant shall be thoroughly coated with P-2, grade 2 preservative.

5.5.1 .12 Gear cases. Fill the gear cases with normal lubricating oil. Shift gears into all possible positions while operating the machine, then close and seal all openings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.

5.5. 1.13 Sliding or friction surfaces. Surfaces of ways, gibs, driving gears, traversing and adjusting screws, and other critical, external operating machined surfaces shall be coated with P-2, grade 2 preservative compound and covered with greaseproof barrier material. Parts of the machine which are adjustable shall be locked into place.

5.5.1.14 Bearings and journals. Grease lubricated bearings shall be preserved with P-11 preservative. Oil lubricated journals shall be filled with lubricating oil P-13, type I, grade optional.

5.5.1.1: Drive belts and pulleys. Drive belts shall either be removed from the machine or released from tension. Ferrous metal pulleys shall be coated with rust-inhibiting primer. Belts removed from the machine shall be packaged in a waterproof, greaseproof bag and sealed.

5.5.1.16 Electrical wiring system. Exposed ends of wires, plug openings, sockets, connector plugs, terminals, and openings into switches and junction boxes, etc. shall be sealed with waterproof tape or cap-plugs, respectively. Electric cable shall be coiled to a minimum safe diameter and secure with cotton tape or twine to prevent uncoiling.

5.5.1.17 Closed dial indicators. Closed dial indicators shall be covered with 1 inch thick uncompressed cushioning material and secured in place with waterproof, pressure sensitive tape. Cushioning materials and tape used shall be water-resistant, non-dusting and non-corrosive.

5.5.1.13 Drawers and door assemblies. Sliding surfaces of drawer guides shall be coated with P-11 preservative grease. Interior surfaces of door hinges shall be lubricated by application of P-10, grade 30 preservative oil. In addition to mechanical locks or catches incorporated in the machine, secure doors and drawers with metal strapping or other devices which are suitable for the purpose intended. Fiberboard pads shall be placed under metal strapping to prevent marring the surfaces.

5.5.1.19 Closure of openings. Small openings, except vents and louvres installed for ventilation purposed, shall be sealed with waterproof, pressure sensitive tape. Large openings shall be covered with waterproof barrier materials. Secure materials in place with tape specified above. When very large openings are covered, or when location of openings render coverings vulnerable to puncture, coverings shall be protected by applying wood, plywood, sheet metal or fiberboard over the materials. Open ends of all piping and fittings shall be plugged to prevent the entrance of foreign materials, using pipe fitting, waterproof tape or plastic cap-plugs.

5.5.1.20 Technical data. Technical data furnished with the machine shall be packaged in a waterproof, greaseproof barrier material and sealed. The package shall be legibly and clearly marked "TECHNICAL DATA" in bold letters. The package shall be protected against puncture and abrasion. The data shall be either attached to the machine with tape, placed in a stowage compartment of the machine, or positioned in a consolidated box containing accessories or disassembled components.

5.5.1.21 Consolidated packaging. Packaged accessories, maintenance tools, and small disassembled components shall be consolidated and placed in drawers and stowage compartments of the machine. When it is not feasible to stow parts as specified above, place parts in weather-resistant fiberboard box or boxes of sufficient strength to handle weight inside. Large bulky disassembled components shall be packed as specified (see 5.6.1.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.

5.5. 1.22 Cushioning. Cushioning materials shall be water resistant, non-dusting and non-corrosive and of sufficient thickness to prevent damage, not less than 1/2 inch thick wrap.

5.5.1.23 Protection of preservatives. All exposed preserved surfaces of machines which are subject to being disturbed during the packing operation shall be loosely covered with greaseproof' barrier materials. Not less than two layers of similar-type barrier material shall be used to separate wood or metal blocking and finished metal surfaces of the machine. Secure barrier in place with waterproof, pressure sensitive tape.

5.5.1.24 Movable parts. Operating heads, sliding tables, rams, counter-balances, and other movable parts shall be secured by mechanical devices incorporated in the machine. The devices alone do not always provide adequate support to prevent movement in the handling and transporting of the item and therefore must be augmented with appropriate wood blocking, metal tie rods used in conjunction with wood blocking, metal strapping, or other devices which will immobilize the part. Handwheels, cranks, levers, and similar controls shall be secured in position with common annealed wire or steel strapping.

5.5.1.25 Positioning assemblies. Assemblies having vertical movement shall be blocked and braced in a lowered position; those capable of moving traversely shall be centered, blocked and braced to prevent movement.

5.5.1.26 Tables. All tables and other components designed for movement shall be removed, or blocked, and all components treated in such a manner that neither the way(s), surfaces(s), nor the anti-friction devices will be subject to brinelling or other damage.

5.5.2 Level C. Level C preservation shall be in accordance with the requirements of MIL-STD-1190.

5.6 Packing. Packing shall be Level A or C, as specified (see 6.2.1).

5.6.1 Level A.

5.6.1.1 Machinery. Machines shall be packed individually in crates conforming to MIL-C-104, type I, class 2, style A. Consolidated containers shall be packed with the machine in the specified crate. Arrangement of contents, anchoring, blocking, bracing, and cushioning shall be in accordance with the appendix to the crate specification, MIL-STD-1186, and as specified herein. Consolidated containers for large bulky disassembled items (see 5.5.1.21) shall conform to PPP-B-601, domestic, style optional.

5.6.1.2 Bolting. Machines shall be secured to the skid base by not less than four bolts. When sufficient bolting points are not available, J-clamps or tie rod assembles, or other devices, shall be used to provide the minimum of four holding points. Bolts shall pass completely through the skid runner or sleeper, as applicable, and the base flooring.

5.6. 1.3 Consolidated containers. Consolidated containers shall be placed inside the shipping container housing the machine. The boxes shall be restrained by use of metal strapping or tie rod assemblies.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:27Z
Check the source to verify that this is the current version before use.

5.6.1.4 Operational oils. operational oils furnished with the machine, but not placed in the machine stowage receptacles, shall be stowed in metal drum(s). Drums shall be positioned upright inside the shipping container housing the machine. blocking shall be applied to immobilize the drum(s) inside the shipping container.

5.6.2 Level C. Level C packing shall be in accordance with the requirements of MIL-STD-1190.

5.7 Marking. All marking shall be in accordance with MIL-STD-129. Bar coding is also required in accordance with MIL-STD-129.

5.8 Workmanship. Workmanship shall be such that when proper procedure is followed, materials and equipment being processed will be provided the maximum protection against corrosion and deterioration consistent with the shipment hazards known or anticipated, and suitable for long term storage.

6. NOTES

6.1 Intended use. The lathe covered by this specification is intended for use in maintenance and field service shops to assist in the overhaul and repair of automotive generators and starter motors.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1.1 and 2.2).
(c) First article when required (see 3.1).
(d) Measuring and indicating device graduation(s) required (see 3.2.2).
(e) Exceptions and additional safety/health requirements when required (see 3.4.7 and 6.5).
(f) Sub-base when required (see 3.6.1).
(g) Drive if different (see 3.6.8).
(h) Type of undercutter drive required (see 3.6.9).
(i) Power supply cord if different (see 3.7.2).
(j) Armature drive motor if different (see 3.7.3).
(k) Voltage if different (see 3.7.4).
(1) Accessories if different (see 3.9).
(m) Painting and finishing if different (see 3.10).
(n) product identification if different (see 3.11).
(0) First article inspection if different (see 4.3).
(p) Quality conformance inspection if different (see 4.4).
(q) Inspection level and AQL for turning and undercutting tests (see 4.6.3 and 4.6.4).
(r) Level of preservation and packing if different (see 5.5 and 5.6).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provision of the Federal Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test and approval of the first article.

6.5 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.6 Sampling. Sampling for quality conformance inspection should be in accordance with MIL-STD-105, using Inspection Levels and Acceptable Quality Levels (AQL) as listed, unless 100 percent inspection is specified.

Inspection or Test	Inspection Level	AQL
Product examination (4.5)	II	l.0
operational test (4.6.1)	100%	---
Alignment test (4.6.2)	100 %	---
Commutator turning test (4.6.3)	II	1.0
Mica undercutting test (4.6.4)	II	1.0
Packaging inspection (4.7)	S-4	4.0

6.7 Subject term (key word) listing.

Mica undercut
Commutator
Armature
Lathe

6.8 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - AL
Air Force - 99

Review activity:
Air Force - 82

Preparing activity:
Army - AL

project NO. 49404588

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-L-14576 - Revision F, Lathe, Armature and Undercutter

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:27Z
Check the source to verify that this is the current version before use.